



AI Recruitment Assistant Platform

Product Requirements Document

Version 1.0 • 2025

Status	Owner	Version	Last Updated
Draft	Product Team	1.0	2025

Executive Summary

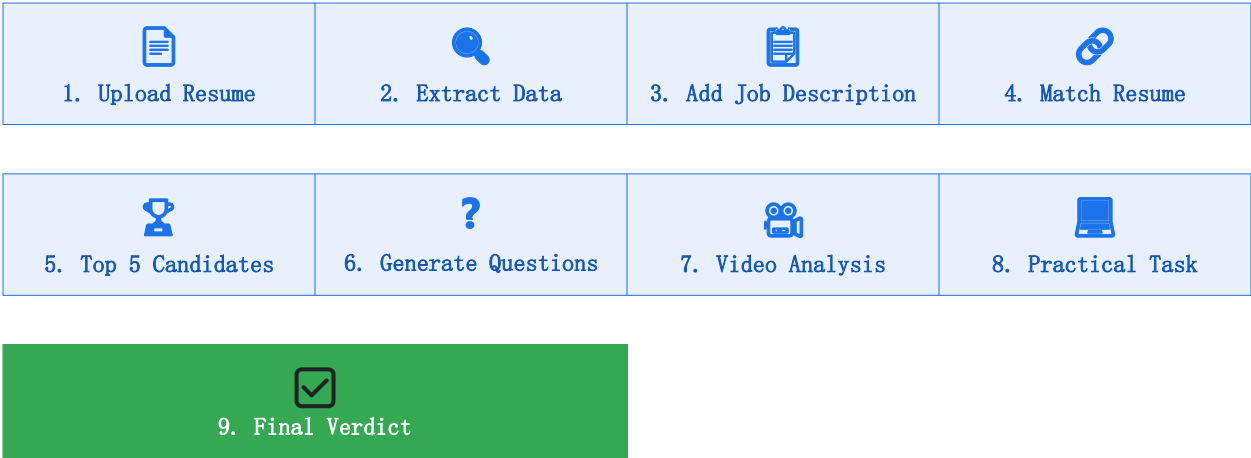
The AI Recruitment Assistant Platform is an end-to-end intelligent hiring system designed to automate, accelerate, and improve the quality of candidate evaluation. The platform covers every stage of recruitment — from resume ingestion to final hiring verdict — using AI-driven analysis at each step.

Key capabilities include:

- Automated resume parsing and structured data extraction from PDF, DOCX, and DOC formats
- Intelligent job description matching with weighted scoring algorithms
- AI-generated interview questions tailored per candidate
- Video interview analysis covering speech, behavior, and authenticity signals
- Practical assignment generation and AI-powered code review
- Consolidated final verdict report for hiring decisions

End-to-End Recruitment Flow

The platform follows a structured 9-phase pipeline. Each phase feeds directly into the next, ensuring a consistent and auditable hiring process.



PHASE 1

Resume Upload & Processing

Ingest candidate resumes in multiple formats for downstream analysis

1.1 Supported File Formats

Format	Extension	Notes
PDF	.pdf	Primary format — full text & metadata extraction
Microsoft Word	.docx	Modern Word format — full fidelity
Legacy Word	.doc	Converted to .docx before processing

1.2 Upload Options

Single Resume Upload

User uploads one resume at a time. Suitable for reviewing an individual candidate or adding a new applicant to an existing pipeline.

- Immediate extraction on upload
- Results displayed inline
- Supports all three file formats

Bulk Resume Upload

User uploads multiple resumes simultaneously. Ideal for high-volume hiring campaigns where speed and batch processing are essential.

- Parallel processing queue
- Progress indicator per file
- Consolidated results dashboard

PHASE 2

Data Extraction

AI parses resumes and structures candidate information into standardised fields

2.1 Personal Information

Extracted Fields

- Full Name
- Email Address
- Phone Number
- Current Location
- Current Company
- Current Designation

2.2 Professional Experience

For each employment entry, the system extracts:

Field	Description
Company Name	Employer organisation name
Job Title	Role / designation held
Employment Type	Full-time, Part-time, Contract, Freelance
Start Date / End Date	Tenure boundaries
Total Duration	Auto-calculated from dates
Responsibilities	Key duties and deliverables
Technologies Used	Tools, languages, frameworks mentioned

Experience Calculations

Metric	Definition
Total Experience	Sum of all employment tenures
Relevant Experience	Experience in domains matching the job description
Domain Experience	Industry-specific depth (e.g., FinTech, Healthcare)

2.3 Education Details

Extracted Fields

- Degree & Course
- Specialisation
- College / University
- Start Year & End Year
- Grade / CGPA

2.4 Skills

Technical Skills

- Programming languages (Java, Python, Node.js)
- Frameworks (React, Spring Boot)
- Cloud & DevOps (AWS, Docker, Kubernetes)
- Databases (MySQL, MongoDB, PostgreSQL)

Soft Skills

- Communication & Presentation
- Leadership & Team Management
- Problem Solving & Critical Thinking
- Conflict Resolution & Ownership

2.5 Additional Extraction

Category	Details Extracted
Certifications	Certificate Name, Issuing Organisation, Completion Date
Projects	Project Name, Role, Duration, Description, Technologies
Social Profiles	LinkedIn, GitHub, StackOverflow, Portfolio
Achievements	Awards, Publications, Patents, Languages spoken

PHASE 3

Red Flag Detection & Job Description Input

Identify hiring risks and capture role requirements

3.1 Resume Red Flag Detection

The AI automatically identifies patterns that indicate potential hiring risks. Flags are used as penalty inputs in the scoring engine.

Red Flag Type	Detection Criteria	Penalty
Employment Gap	Any gap > 3 months between roles is flagged with start date, end date, and duration	-5% to -15%
Education Gap	Delayed graduation or unusual academic breaks detected	-5%
Job Hopping	Multiple tenures under 12 months or > 3 job changes in 5 years	-10%
Missing Contact Info	No email, phone number, or LinkedIn profile present	-5%
Missing Experience Dates	Employment entries lack start or end dates	-5%

3.2 Job Description Input

Recruiters provide role requirements through a structured input form. This data drives the matching engine in Phase 4.

Input Field	Description	Status
Job Title	Official role name (e.g., Senior Backend Engineer)	Required
Job Description	Full narrative description of responsibilities	Required
Required Skills	Must-have technical skills (e.g., Java, Spring Boot, Microservices)	Required
Preferred Skills	Nice-to-have skills (e.g., AWS,	Optional

Input Field	Description	Status
	Kubernetes)	
Experience Range	Minimum and maximum years of relevant experience	Required
Education Requirements	Minimum degree or qualification expected	Optional
Domain Requirements	Industry context (e.g., FinTech, Healthcare, E-commerce)	Optional

PHASE 4

Resume Matching Engine

Compare every uploaded resume against the job description using weighted scoring

4.1 Candidate Scoring Formula

Each candidate receives a composite score calculated from the following weighted components:

Component	Description	Weight
Skills Match	Required & preferred skills overlap	35%
Experience Match	Total and relevant years of experience	25%
Project Relevance	Technology stack, domain, and project complexity	15%
Certification Relevance	Industry-recognised certifications aligned to the role	10%
Education Match	Degree level and field of study	10%
Red Flag Penalty	Deductions for gaps, hopping, and missing data	-5% to -20%

Formula

- Final Score = (Skills × 35%) + (Experience × 25%) + (Projects × 15%) + (Certifications × 10%) + (Education × 10%) - Red Flag Penalty

- Score Range: 0 - 100

4.2 Candidate Analysis Output

For each candidate, the system generates a structured analysis profile:

- Match Percentage — Overall fit against the job description (e.g., 87%)
 - Matching Skills — Skills the candidate possesses that align with requirements
 - Missing Skills — Required skills absent from the resume
 - Strengths — Narrative summary of strong areas (e.g., large-scale project experience)
 - Weaknesses — Areas where the candidate falls short
 - Red Flags — Flagged concerns with context (e.g., "8-month employment gap in 2022")
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PHASE 5

Top 5 Candidate Selection

Rank and surface the highest-scoring candidates for interview

5.1 Selection Criteria

The system selects the top 5 candidates ranked by their composite score. Selection considers all four matching dimensions holistically.

- Overall composite score (primary sort key)
- Skill match percentage as a tiebreaker
- Red flag severity as a secondary filter
- Experience alignment with the stated experience range

5.2 Candidate Ranking Output

Rank	Candidate	Score	Top Matching Skills	Status
1	Candidate A	92 / 100	Java, Spring Boot, AWS, Docker	Proceed to Interview
2	Candidate B	88 / 100	Java, Microservices, Kubernetes	Proceed to Interview
3	Candidate C	85 / 100	Node.js, React, AWS	Proceed to Interview
4	Candidate D	82 / 100	Python, Django, PostgreSQL	Proceed to Interview
5	Candidate E	80 / 100	Java, Spring Boot, MySQL	Proceed to Interview

PHASE 6

AI Interview Question Generation

Generate personalised question sets for each shortlisted candidate

6.1 Technical Questions

The AI generates 10 technical questions per candidate, calibrated to their skill profile and experience level.

Generation Inputs	Difficulty Distribution		
	Level		Count
	Candidate's existing skill set	Easy	2 questions
		Medium	5 questions
		Hard	3 questions
Missing or weak skills identified during matching			
Specific job description requirements			
Years of experience (difficulty calibration)			

6.2 Behavioural Questions

5 behavioural questions are generated per candidate using the STAR-method framework, targeting the following competencies:

Competency	Focus Area
Leadership	How the candidate has led teams or initiatives
Team Collaboration	Experience working within cross-functional teams
Problem Solving	Approach to ambiguous or high-pressure situations
Conflict Resolution	Handling disagreements or competing priorities
Ownership & Accountability	Taking responsibility for outcomes

PHASE 7

Video Interview Analysis

Upload recorded interview sessions for AI-driven multi-dimensional evaluation

7.1 Video Upload

After the interview session, the recording is uploaded to the platform for automated analysis.

Supported Formats

- MP4 (.mp4)
- QuickTime (.mov)
- AVI (.avi)
- WebM (.webm)

7.2 Analysis Components

Speech-to-Text & Transcript

The full interview is transcribed. The system produces a structured question - answer mapping with timestamps, enabling downstream quality analysis.

Answer Quality Analysis

Dimension	Evaluation Criteria
Technical Accuracy	Correctness and depth of technical responses
Relevance	How directly the answer addresses the question asked
Completeness	Coverage of key concepts expected in a strong answer
Real-World Experience	Presence of concrete examples from past work

Communication Analysis (Weight: 20%)

Metric	Description
Speaking Clarity	Articulation, pronunciation, and sentence structure
Grammar & Vocabulary	Language quality and professional register
Confidence Level	Assertiveness and pacing of speech
Filler Word Count	Frequency of "umm", "ahh", "like", "you know"
Response Clarity	Conciseness and logical flow of answers

Confidence & Engagement Analysis (Weight: 10%)

Signals Evaluated

- Eye contact and sustained gaze toward camera
- Facial expressions and micro-expressions
- Voice stability and variation
- Hesitation frequency and duration
- Overall engagement level

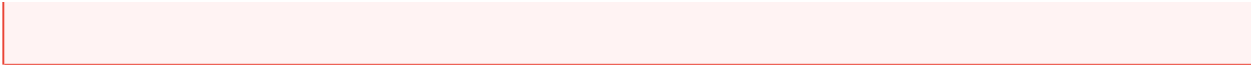
⚠ Important: Confidence score is treated as an indicator only. It must not be used as a standalone hiring decision factor, as confidence signals vary significantly across individuals and cultural backgrounds.

Authenticity & Cheating Risk Detection

Risk Level	Description
Low Risk	No suspicious activity detected — interview proceeds normally
Medium Risk	Some unusual off-camera behaviour — flagged for human review
High Risk	Multiple suspicious patterns detected — significant concern raised

Suspicious Behaviour Signals

- Frequent off-camera eye movement (reading from another screen)
- Multiple faces detected in frame
- Candidate leaving the camera view repeatedly
- Voice mismatch or audio anomalies
- Face mismatch across different segments



7.3 Interview Scoring Model

Category	Weight	Description
Technical Knowledge	40%	Accuracy, depth, and breadth of answers
Answer Quality	20%	Relevance, completeness, and examples provided
Communication Skills	20%	Clarity, vocabulary, filler-word rate
Confidence Score	10%	Engagement, eye contact, voice stability
Behavioural Assessment	10%	STAR responses, teamwork, ownership

Interview Qualification Rule

- Score $\geq 75 \rightarrow$ Candidate proceeds to Practical Assessment (Phase 8)
- Score $< 75 \rightarrow$ Candidate is rejected or flagged for manual HR review

PHASE 8

Practical Task Generation

Create a bespoke technical assignment for each qualifying candidate

8.1 Assignment Parameters

Parameter	Value	Notes
Duration	3 Hours	Fixed time limit for all candidates
Customisation	Per Candidate	Tailored to job description, score, and skill gaps
Difficulty	Experience-Calibrated	Adjusted based on seniority level

8.2 Task Components

Component	Description	Applicability
Coding Task	Implement a functional feature or REST API endpoint	Core requirement
Database Task	Design and implement a normalised database schema	Core requirement
Architecture Task	Produce a high-level system design diagram with rationale	Senior roles
Documentation Task	Write a README and API documentation	All roles

PHASE 9

Practical Submission Analysis

AI reviews submitted code and produces a detailed quality assessment

9.1 Submission Formats

Accepted Submission Types

- ZIP archive — full project directory compressed
- GitHub Repository — public or private repo URL with access
- Source Code Folder — direct upload of unpacked source files

9.2 Code Review Dimensions

Dimension	What Is Evaluated	Weight
Code Quality	Clean code principles, naming conventions, readability, maintainability	20%
Architecture	Design patterns, separation of concerns, scalability decisions	15%
Functionality	Requirement coverage, business logic correctness, edge case handling	40%
Security	Authentication, authorisation, input validation, injection prevention	10%
Testing	Unit test coverage, integration tests, test quality	10%
Documentation	README completeness, API docs, inline code comments	5%

Practical Score Formula

- Practical Score = (Functionality × 40%) + (Code Quality × 20%) + (Architecture × 15%) + (Security × 10%) + (Testing × 10%) + (Documentation × 5%)
- Score Range: 0 - 100

9.3 Practical Assessment Output

The AI generates a structured report covering:

- Score — Numerical grade out of 100 (e.g., 84/100)
 - Strengths — Specific technical merits identified (e.g., clean architecture, thorough API design)
 - Missing Areas — Requirements not fulfilled or quality gaps (e.g., no unit tests, missing input validation)
 - Recommendations — Actionable improvement suggestions
 - Executive Summary — Plain-language hiring recommendation narrative
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FINAL VERDICT

Consolidated Hiring Decision

Combining all phase scores into a single hiring recommendation

Final Score Composition

Component	Weight	Description
Resume Matching Score	25%	Weighted skill, experience, and education fit
Interview Assessment Score	40%	Technical, communication, and behavioural evaluation
Practical Task Score	35%	Code quality, functionality, and architecture

Final Score Formula

- Final Score = (Resume Score × 25%) + (Interview Score × 40%) + (Practical Score × 35%)
- Score Range: 0 - 100

Verdict Thresholds

Score Range	Verdict	Action
85 - 100	Strong Hire	Candidate is highly recommended — fast-track offer
70 - 84	Hire	Candidate meets requirements — proceed with offer
55 - 69	Maybe / Manual Review	Borderline candidate — escalate to senior panel
0 - 54	Reject	Candidate does not meet the minimum bar

Final Report Contents

For each candidate who completes the full pipeline, the platform generates a downloadable PDF report containing:

- Candidate profile summary (personal info, experience, education)
- Resume match score with skills breakdown
- Identified red flags and their impact on scoring
- Interview assessment — per-question scores and overall analysis
- Communication and confidence review
- Authenticity and cheating risk assessment
- Practical task score with strengths, gaps, and code review notes
- Final composite score and hiring verdict
- Hiring manager recommendation narrative

Appendix — Score Summary Reference

Phase	Score Name	Range	Gate Condition
Phase 4	Resume Match Score	0 - 100	Proceeds if top 5 rank
Phase 7	Interview Score	0 - 100	≥ 75 required to advance
Phase 9	Practical Score	0 - 100	Included in final composite
Final	Composite Score	0 - 100	≥ 70 = Hire recommendation

AI Recruitment Assistant Platform — PRD v1.0
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